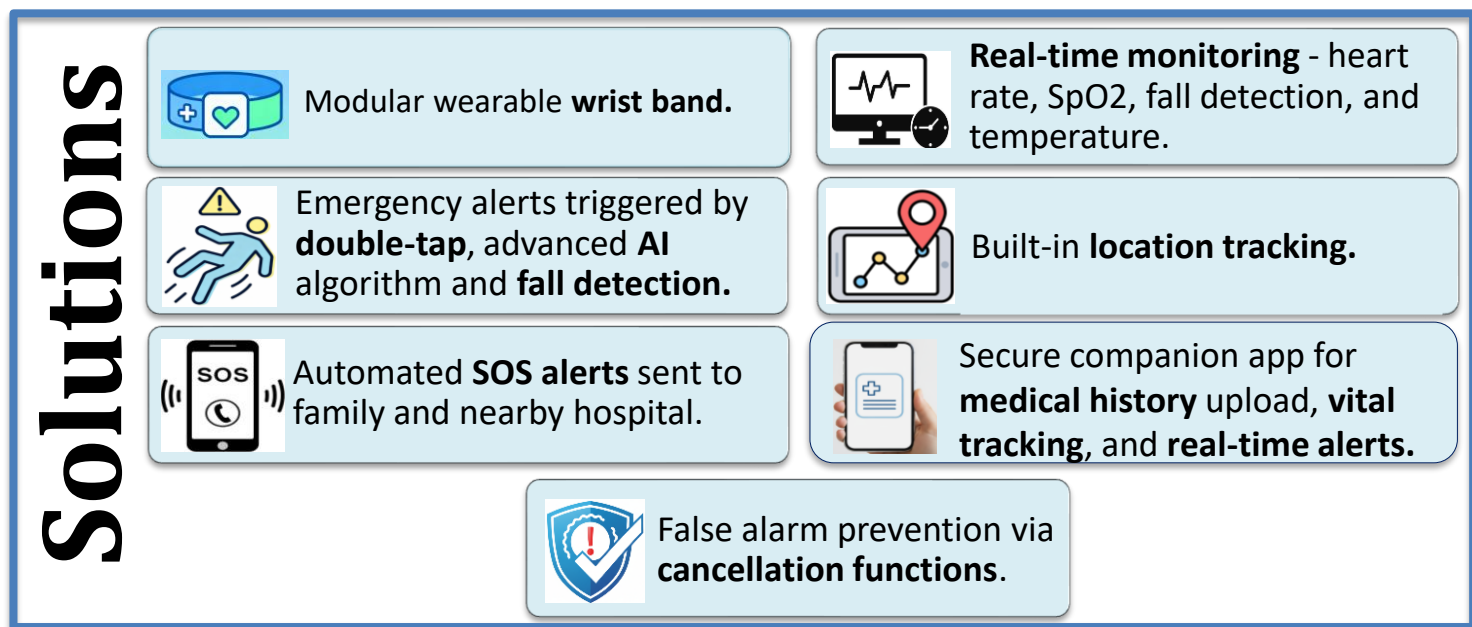
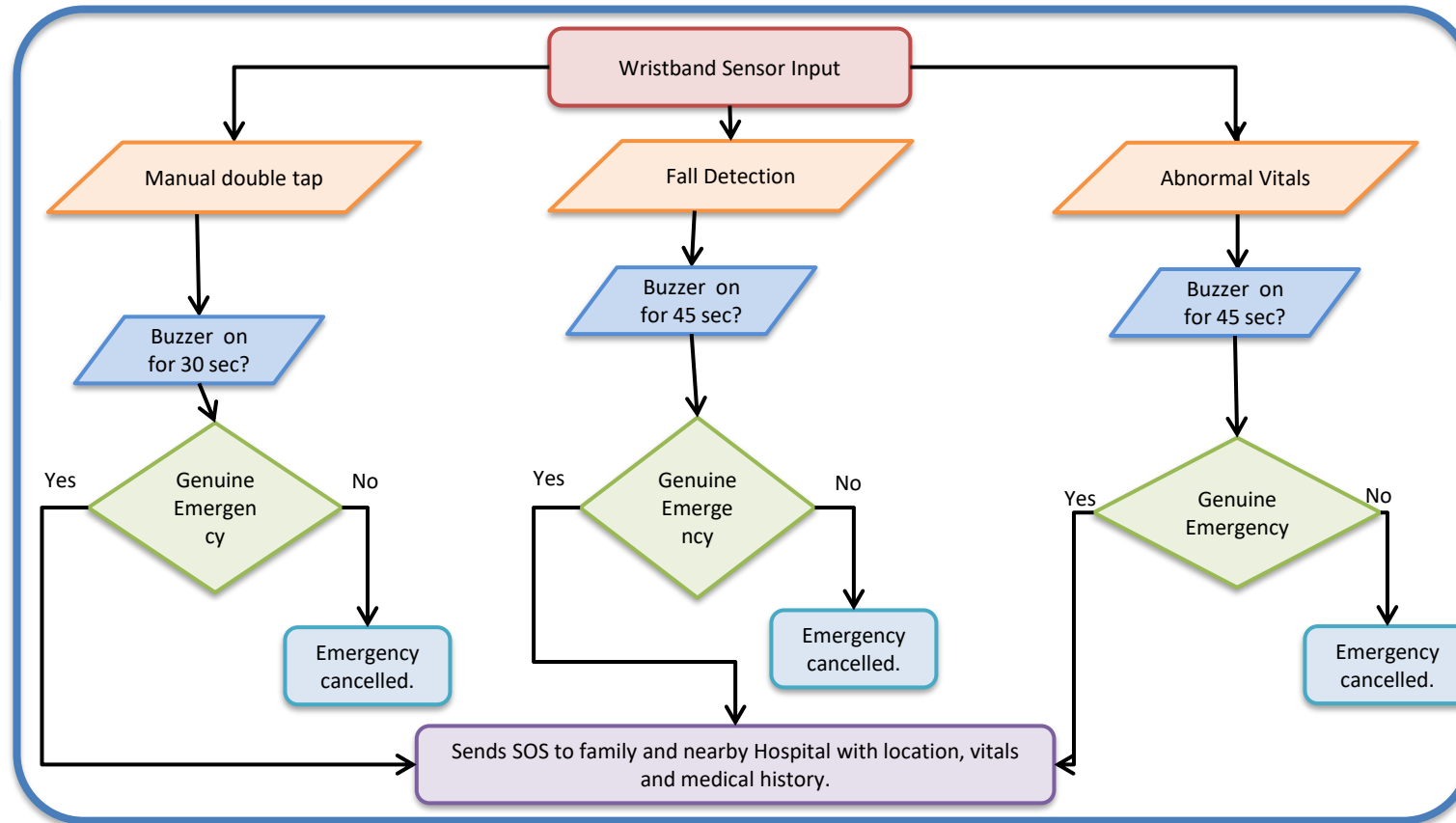




- ### Uniqueness
- ✓ All-in-one device: SOS, vitals, fall detection, location sharing.
 - ✓ Two-step emergency verification with double tap and cancel to reduce false alarms.
 - ✓ Auto-share health record, vitals, and location for rapid response.
 - ✓ Secure wireless health insights with cloud support.

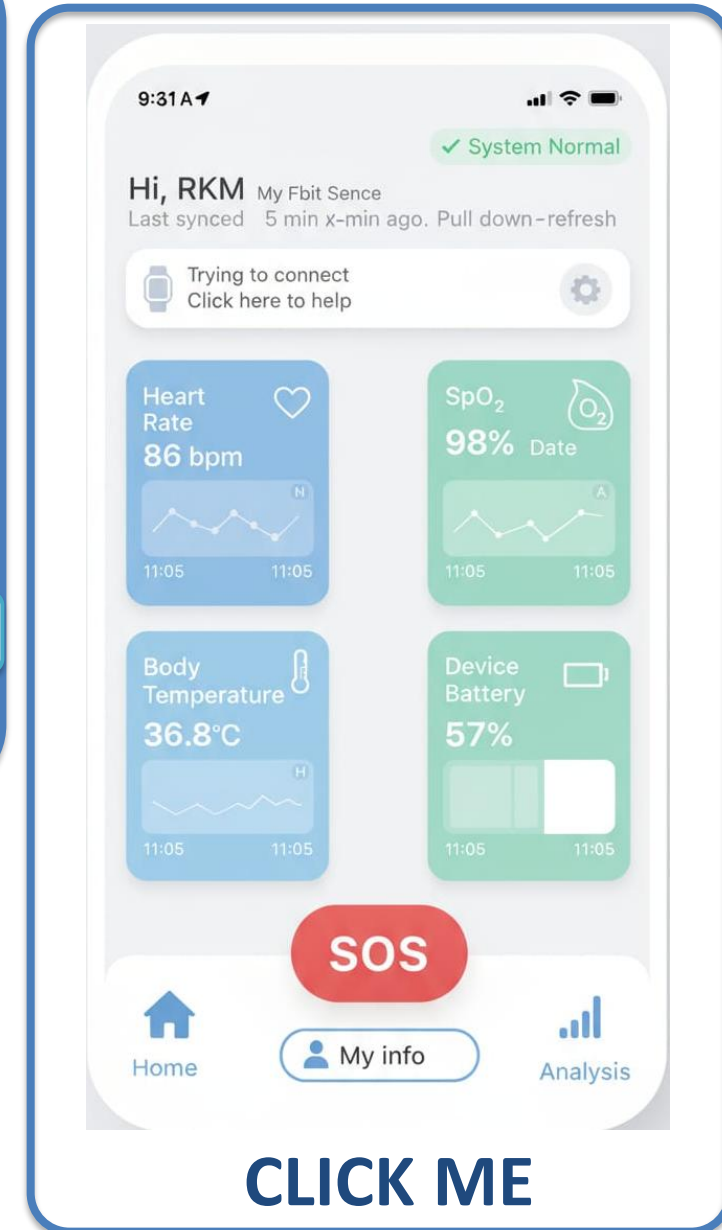
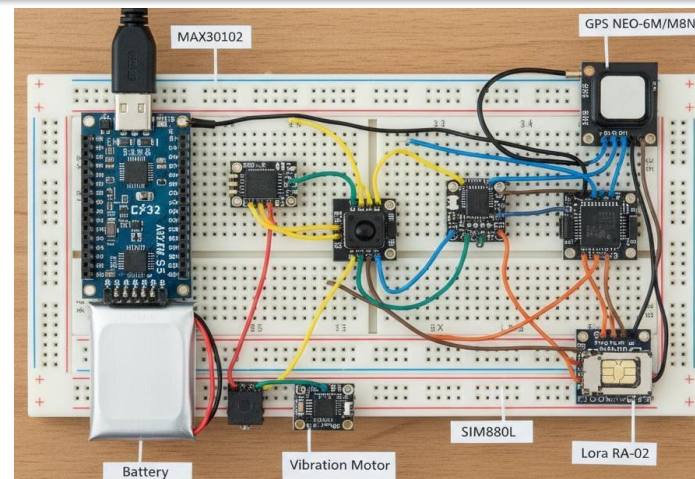


TECHNICAL APPROACH



CONTROLLER & SENSORS

- 1) ESP32
- 2) SIM800L GSM module
- 3) MAX30102
- 4) MPU6050
- 5) GPS Module (NEO-6M)



CLICK ME

Feasibility of Idea

- High demand in elderly care and remote monitoring.
- Medical history and medication integration improves emergency response.
- Algorithms enables early detection.

Challenges and Risks

- High R&D and advanced algorithms needed.
- Avoiding false alarms and ensuring data privacy.
- Limited clinical diagnostics from wearable sensors.
- Connectivity and battery limitations.

Strategies to Overcome

- Consulting healthcare professionals for validation.
- Multi-step alarm verification to reduce false alerts.
- Use secure, encrypted data handling.
- Thorough real-world testing.
- Manual false alarm cancellation button.

POTENTIAL IMPACTS



Reduces emergency response delays with instant alerts



Automated connectivity with caregivers and doctors



Empowers vulnerable user independence and confidence



Provides immediate help even if unconscious or speech impaired



Continuous monitoring of vitals and location



Enhances peace of mind for families and caregivers

Social Benefit

- ❑ Promotes inclusive healthcare for all users
- ❑ Improves quality of life for seniors and chronically ill
- ❑ Strengthens social safety nets

Economic Benefits

- ❑ Enables remote health trend analysis
- ❑ Affordable, scalable, and energy-efficient design

Environmental Benefit

- ❑ Minimizes environment impact
- ❑ Durable, upgradable hardware
- ❑ Extends lifespan and reduces e-waste



RESEARCH AND REFERENCES



Research

Nearly 75% of elderly Indians have at least one chronic disease, with 24% experiencing at least one activity of daily living (ADL) limitation that can restrict their ability to call for help in emergencies



References

A) Read research paper

1. Rahman, S., Walker, D., & Sultan, P. (2017). Medical identification or alert jewellery: an opportunity to save lives or an unreliable hindrance? *Anaesthesia*, 72(9), 1139–1145.
<https://doi.org/10.1111/anae.13958>
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2. Kumar, A., Sharma, R., & Gupta, P. (2022). *Smart Band: An Integrated Device for Emergency Management*. *International Journal of Advanced Research in Engineering and Technology*, 13(4), 156–162. Retrieved from
https://www.researchgate.net/publication/353567400_Smart_Band_An_Integrated_Device_for_Emergency_Management
3. Tabrez, S., Azam, F., & Hassan, M. (2018). *Towards a fully automated bracelet for health emergency solution*. *Procedia Computer Science*, 132, 123–130. <https://doi.org/10.1016/j.procs.2018.05.257>

B) Viewed already existing solution.

1. <https://share.google/IK8Ct8ki64AGTWfME>
2. <https://www.minew.com/product/b7-button-wristband/>

C) Visted LG multi-specialty hospital, Maninagar, Ahmedabad, Gujarat.

D) Reviewed idea from 4 -5 Residential doctors.